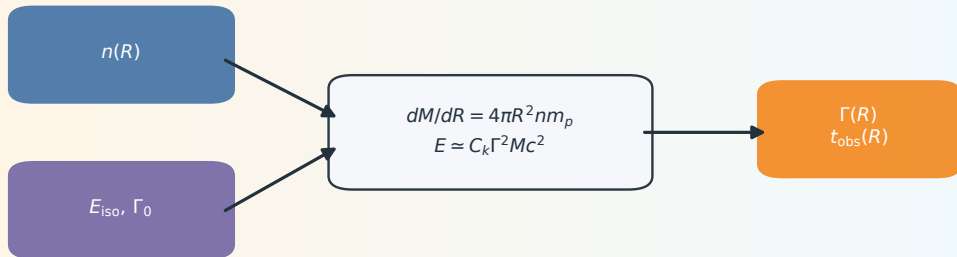


Forward-shock dynamics

Swept-up mass converts initial ejecta energy into decelerating blast-wave motion.



$$\Gamma \propto R^{-(3-k)/2}, \quad \Gamma \propto t_{\text{obs}}^{-(3-k)/2(4-k)}$$